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Title (required)

Predicting PET amyloid in prodromal Alzheimer's disease using Weighted Hidden Markov Model

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Introduction (required)

Quantifying amyloid proteins through PET imaging is the gold standard for assessing Alzheimer's disease (AD). Compared with PET scans, fMRI offers a noninvasive, affordable alternative that can capture dynamic interactions within whole-brain neural networks. Existing work shows relationships between structural MRI and CSF biomarkers,^1 and studies have also identified Tau-dependent dynamic Functional Connectivity (dFC) in animal experiments.^2 Compared with deep learning, Hidden Markov Models (HMMs) offer a more interpretable form of dFC. However, modulating continuously changing variables through HMMs is mathematically challenging. In this abstract, our primary goal is to investigate the relationships between PET amyloid concentrations in prodromal AD using HMMs to link PET concentrations with dFC measures from resting-state fMRI data.^3 We present a Weighted Hidden Markov Model (WHMM) that modulates the amyloid-dependent term using a Gaussian mixture, with all formulas rigorously defined using the Expectation–Maximization (EM) algorithm. The proposed model is applied to the Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset. Results show that it can successfully characterize amyloid-related changes, yielding a correlation of 0.25 between the real and predicted amyloid values.

Methods (required)

Resting fMRI scans are selected from ADNI phase 2 and phase 3.^4 The selection criteria are TR = 3s, resting fMRI with a time length of 135/192/195, and at least one PET scan performed within 6 months. In total, 1446 scans were selected with the corresponding nearest PET scan results. All subjects underwent preprocessing, including slice-time correction, realignment, and normalization to MNI space. The fMRI data were denoised by regressing out CSF and gray-matter signals and then parcellated into 32 regions using the CONN-32 atlas. For each region, the meantime series was extracted, yielding a 32-dimensional representation. The amyloid value, referred to as the summary SUVR or AV45, was selected. This value was cropped between 0.9 and 1.8 and then rescaled to 0–1. Figure 1(a) presents the distribution of AV45. The data were split into train/eval/test sets with a ratio of 7/2/1, and there was no subject overlapping.

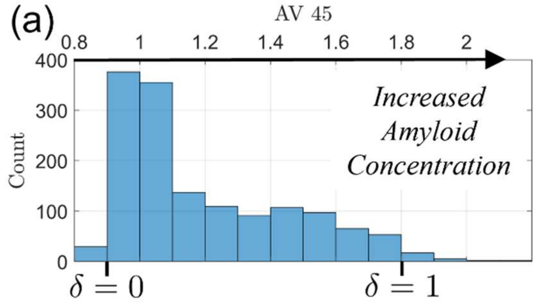
Figure 1(b) shows the HMM defined in this system. fMRI data after preprocessing and application of the CONN-32 mask are reduced to a time series with dimension $T \times p$, where $p = 32$. We assume the system transitions among K states, and the observation at each time point is generated using a Gaussian distribution with mean μ and covariance σ. The covariance σ naturally measures the relationships between different networks, which is referred to as dFC in previous work.

To model the effect of amyloid concentration on the brain, we consider the weighted Gaussian model defined in Figure 1(b-d), where for a subject s, the Gaussian mixture is defined using weight δ_s. Note that δ_s is fixed for this subject across all time points and all hidden states. This differs from the traditional Gaussian mixture HMM, where the mixture ratio is shared across all subjects but not across hidden states. To better illustrate this model, we call it the Weighted Hidden Markov Model (WHMM). From an interpretational point of view, μ_1, σ_1 corresponds to amyloid neutral, and μ_2, σ_2 correspond to amyloid elevated. We note that a similar approach is proposed to characterize feature weights,^5 but δ is determined using only an unsupervised approach.

The parameter extraction shown in Figure 1(e) is divided into two parts. In the generative part, we implement the EM algorithm on the training data to estimate two sets of Gaussians with a given mixture ratio δ. Once the model converges, the Gaussians are fixed, and the ratios are updated using the formula proposed in Figure 1(e). The train, evaluation, and test accuracy can then be derived. Starting from the results of the generative model, the discriminative model performs supervised training and minimizes the Mean Square Error (MSE) loss between the predicted mixture ratio and the true mixture ratio. The MSE is monitored on the evaluation dataset. Finally, parameters that lead to smallest MSE in evaluation dataset are selected.

Methods Figure (Optional)

(a)



(b)

Observation: $x_{st} \in \Re^{1\times p}$ Time length: $T = 135/192/195$

Subject index: s Feature dimension: $p = 32$

Amyloid value: δ_s Hidden states: h $\underbrace{\bigcirc\bigcirc\bigcirc}_{K=3}$

Transition probability: $p(h_t|h_{t-1}) = A_{h_{t-1}h_t}$

State occupational probability: $\gamma_{stk} = p(h_t = k|x_{st})$

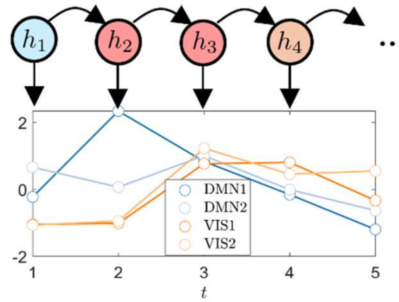
(c) Emission probability

$$p(x_{st}|h_t = k) = (1-\delta_s)P^{(1)}_{skt} + \delta_s P^{(2)}_{skt}$$

$$P^{(i)}_{skt} = (2\pi)^{-p/2}\det(\sigma^{(i)}_k)^{-1/2}\times$$

$$\exp\left[-\frac{1}{2}(x_{st}-\mu^{(i)}_k)^T(\sigma^{(i)}_k)^{-1}(x_{st}-\mu^{(i)}_k)\right]$$

(d)



(e)

Generative	Training	Initialization:	$\{A, \mu^{(1,2)}, \sigma^{(1,2)}\}$
		Fixed :	$\{\delta_{\text{train}} = \delta_{\text{train (real)}}\}$
		Update (until convergence) :	$\{A, \mu^{(1,2)}, \sigma^{(1,2)}\}$
	Testing	Initialization:	$\{\delta_{\text{train}} = \delta_{\text{train (real)}}, \delta_{\text{test}} = 0.5\}$
		Fixed :	$\{A, \mu^{(1,2)}, \sigma^{(1,2)}\}$
		Update (until convergence) :	$\{\delta_{\text{train}}, \delta_{\text{test}}\}$ $\delta_s = \sum_{tk} \frac{\gamma_{stk}}{T} \frac{\delta_s P^{(2)}_{stk}}{(1-\delta_s)P^{(1)}_{stk} + \delta_s P^{(2)}_{stk}}$
Discriminative	Initialization using generative training results		
	Optimizer: adam	Implementation: PyTorch	
	Learning rate: 1e-4	Loss: $\mathcal{L} = (\delta_{\text{predict}} - \delta_{\text{real}})^2$	

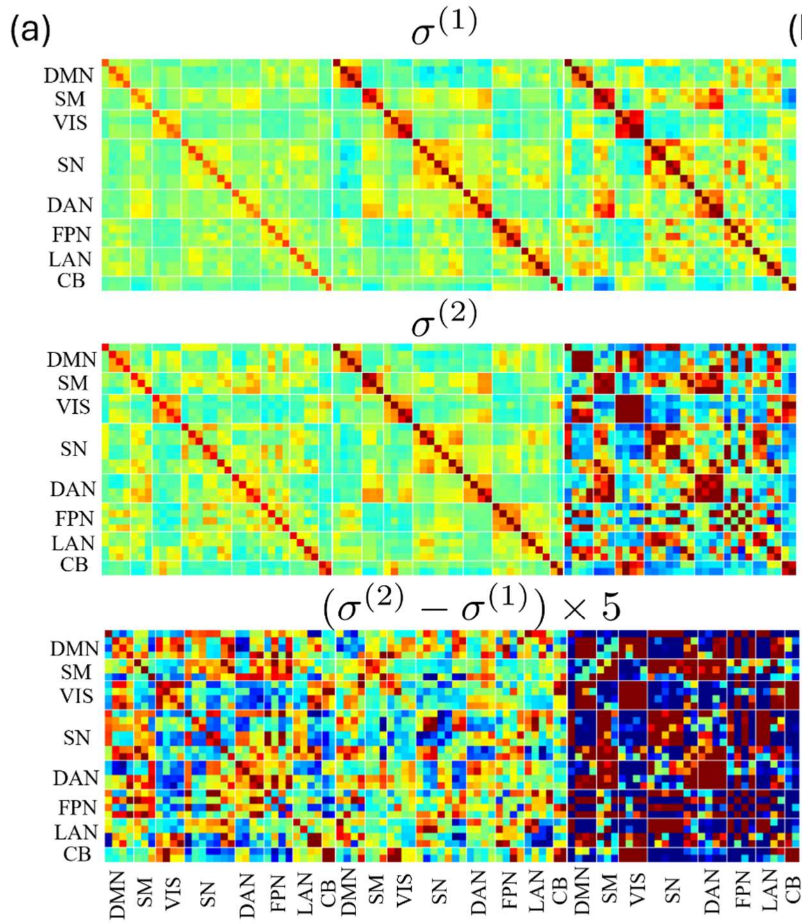
Figure 1: (a) Distribution of amyloid concentration across all subjects, with each subject associated with one normalized value δ_s. (b) Definition of the HMM model. (c) Emission probability for the WHMM proposed in this work. (d) Illustration of the HMM on four networks—two Default Mode Networks (DMNs) and two Visual Networks (VISs)—chosen from the CONN-32 atlas. (e) Generative (EM algorithm) and discriminative (Gradient-based) training steps for the WHMM.

Results (required)

Figure 2(a) presents the results after discriminative training. The hyperparameters for the HMM are *K = 3*, and we use a Mixture of Factor Analyzers (MFA) with reduced dimension *q = 8*. The first row shows the dFC for the three hidden states, representing amyloid-neutral connectivity, while the second row shows the dFC with elevated amyloid. Their difference, shown in the third row, indicates the amyloid effect on dFC. We observe that for the first two hidden states, except for dFC connected to the visual network (VIS) and cerebellum (CB), amyloid concentration generally increases connectivity. However, for VIS and CB, the connectivity decreases.

Figure 2(b) presents the MSE versus training epoch. We observe that the training loss decreases quickly, while the evaluation and testing losses gradually converge after around 1000 steps, suggesting overfitting in the model. In Figure 2(c) and (d), we plot the relationship between the real and predicted AV45 values. To avoid statistical variance, these plots overlay results from 100 different data separations. On average, we observe a training MSE of 0.037, corresponding to a correlation of 0.70, while the testing MSE is 0.065 with a corresponding correlation of 0.25.

Results Figure (Optional)



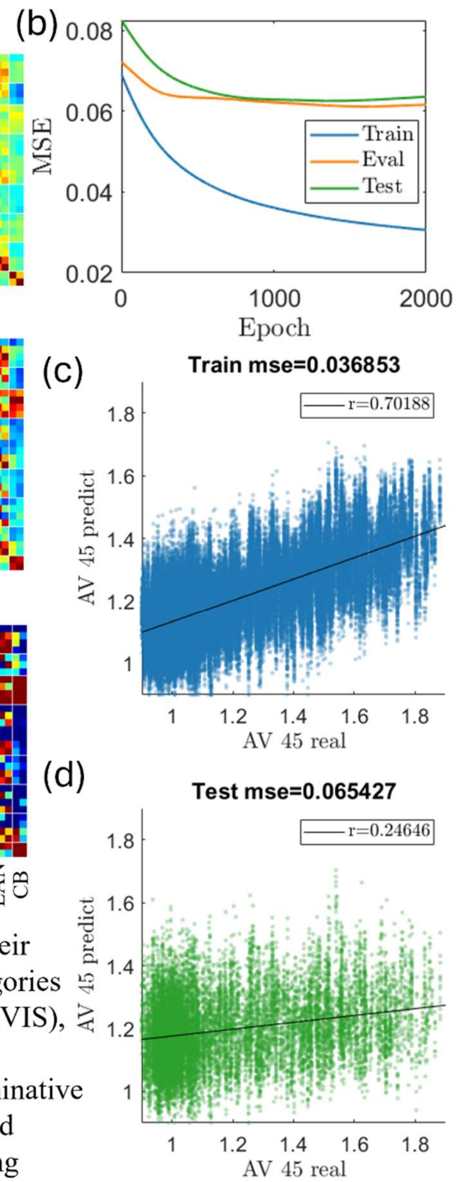


Figure 2: (a) dFC for amyloid-neutral, amyloid-elevated, and their difference (scaled by a factor of five). 32 networks from 8 categories are chosen: Default Mode (DMN), Sensorimotor (SM), Visual (VIS), Salience (SN), Dorsal Attention (DAN), Frontoparietal (FPN), Language (LAN), and Cerebellar (CB). (b) MSE versus discriminative training epoch for one data separation. (c) The real and predicted amyloid values in the training dataset, and (d) those in the testing dataset. Both (c) and (d) show overlays from 100 data separations.

Conclusion (required)

The findings presented in this abstract are threefold. First, we propose a Weighted Hidden Markov Model based on a Gaussian mixture that can represent continuous amyloid values through the mixture ratio. Second, we propose an EM algorithm to solve this problem, with each step analytically defined and requiring no optimization. Third, we demonstrate that the proposed HMM framework has the potential to predict amyloid burden, showing a clear linear relationship between the real and predicted AV45 values in the testing dataset. All results are averaged over 100 random data splits to mitigate statistical variance.

References/Citations (required)

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Acknowledgement

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